

Design Verification Plan & Report

Design Verification Plan and Report (virtual) — VBF-T8R1FLASH-DVPR-01

Case: t8_r1_flash | **Date**: 2026-08-25 | **Design**: V4 Thermal-tuned

#	Item	Condition	Result	Determination	Source
1	1C discharge capacity	1C CC to 2.5 V, 25 °C, DFN	3.452 Ah	meets design 3.40 Ah	cell/r1_v4_1c.json
2	5C discharge retention	5C CC to 2.5 V, 25 °C, DFN	99.0 % (5C 3.418 Ah / 1C 3.452 Ah)	PASS ≥ 90 %	cell/r1_v4_retention.json
3	Energy density	contract formula	473.0 Wh/kg	PASS ≥ 446.18	cell/r1_v4_energy.json
4	Cell mass	contract layer stack	26.8 g	PASS ≤ 40 g	cell/r1_v4_energy.json
5	4C fast-charge temperature rise	4C charge, 45 °C ambient, lumped thermal, DFN	T_max 331.98 K (ΔT 13.8 K)	PASS ≤ 333.15 K	cell/r1_v4_4c.json
6	4C fast-charge plating	same + plating model, min anode potential	0.0247 V > 0	PASS, no plating	cell/r1_v4_4c.json:anode_potential_v
7	Voltage window	parameter set	4.20–2.50 V	meets spec	parameter set
8	Overcharge → thermal runaway (informational)	1C to 2.5 V, 0.5C to 4.7 V; run-tr coupling	T_max 300.57 K; triggered = False	not triggered (informational)	cell/r1_v4_ovc.json, r1_v4_tr.json

N/A items (beyond pure-simulation boundary, require physical experiment): nail penetration, crush, drop, rate-pulse internal resistance, cycle life (standard SEI model artifact — see aging note), –20 °C low-temperature retention.

Conclusion: All contract verification items PASS. Uncovered items listed above (cite as paper limitations).